© TXPMTKEY

KR200445241Y1 20090710

EPO

SUBJECT OF THE INVENTION
Umbrella handle with increasable length.

The present invention relates to an umbrella handle, and more particularly to an umbrella handle capable of increasing the length of the umbrella handle and controlling the use of the umbrella handle safely, wherein the umbrella handle according to the present invention is adjustable in length And safety is increased, and it is applied to unfolding and folding the umbrella automatically by preventing the button being pressed unintentionally during use.

Umbrella is a kind of tool used by people in everyday life and is an indispensable commodity to protect from rain and sunshine. The first umbrella was used as a waterproofing tool and consisted of a middle, a mountain and a mountain, As the industry develops, the umbrella increases the length of the handle so that it is convenient for people to catch, and at the same time, the middle bong is improved and the folded structure of the umbrella is reduced by the polygonal structure, It was convenient and easy to control, however, because people could not satisfy the ever-increasing demand for high-quality living, they developed an umbrella that folded and unfolded automatically in the 90s of the 20th century, Through the automatic control mechanism installed in the umbrella handle Automatic umbrella, especially portable umbrella folding umbrella from the handle of the main parts of the umbrella, except for the internal control of the installation has been constantly changed and improved, but no structural There is no change, it is small, the structure is compact and convenient to carry, but the length is much smaller than the width of the human hand and it is inconvenient to hold it in the hand, and the use of the mouse is not fixed, Friction, collision, etc., and the reliability of necessary safety is lacking because it can cause error manipulation by pressing the button of the handle due to the action of external force during carrying.

Therefore, the object of the present invention is to provide an umbrella handle capable of extending the length of the umbrella handle. The handle of the umbrella is adjustable in length under the control of the user, Another object of the present invention is to improve control within the umbrella handle and to increase the mechanical operating reliability of the umbrella handle length, thereby facilitating assembly and lowering the manufacturing cost. In order to accomplish the above object, the present invention is configured as follows: In the umbrella handle in which a seat inside the umbrella handle is provided and the length of which can be adjusted by operating a control button, And the handle extension gun is fitted to the handle, And a connecting mechanism is provided between the handle extension and the handle to increase the length of the umbrella

handle by realizing docking and positional connection of the handle and the handle. Wherein the protrusion is provided on the outer wall of the handle or on the inner wall of the handle extension gun in the longitudinal direction of the handle and the groove is provided on the inner wall of the handle extension gun or on the outer wall of the handle, Wherein the protrusion is a convex block provided on the inner wall of the handle extension gun, and the groove is an extension groove member provided on the outer wall of the handle, The end of the extending groove member has a flange, And the protrusion of the umbrella handle and the handle extension bar is prevented from being separated from the umbrella handle and the handle extension bar. The elongated groove member can be elastically acted by the elastic member and is easy to assemble. Wherein at least one block and an extending groove member are installed and arranged symmetrically on an axis line of the handle extension gun and the handle to form a stable structure. This design is convenient and easy to control by using the above-mentioned structure to suitably increase the handle length of the automatic umbrella. When carrying or not using it, it is possible to effectively reduce the handle length, reduce the volume, control the umbrella handle The button is covered with an umbrella handle extension, effective outer layer By receiving a call it can prevent error operation caused by external force. In addition, the above-described structure is simple, and processing and assembling are convenient, so that it can be widely applied to an automatic umbrella.

Of the embodiments described below, the umbrella handle, the control button and the control mechanism are all available in various forms, and at the same time, many materials have already reliably depicted the form of the control and control buttons, 1 to 3 are explanatory diagrams of the structure of Embodiment 1 of the present invention. The umbrella handle 10 includes a sheet 13 inside the umbrella handle And the inner seat 13 of the umbrella grip is coupled through the engagement hole 101 of the umbrella handle 10 and the projections 131 distributed on both sides to realize connection and fixing of the two. There is an umbrella pull extension gun 11 and it is possible to partially enclose the umbrella handle 10 to form a protection against the umbrella handle 10 and at the same time operate the control button 12 for convenient control The umbrella handle extension tool 11 includes a bottom cover 113 and the bottom cover 113 is fixed below the umbrella handle extension gun 11 and extends upwards on both sides The projection 114 has two positioning holes 112 on the upper and lower sides to confirm the position of the umbrella handle extension gun 11. On both sides of the umbrella handle 10, A spring 103 is provided in the spring 103 and the spring 103 supports the spring bead 104. In the course of the upward movement of the umbrella handle extension gun 11, The gripper 104 slides up and down along the projection 114 under the support of the spring 103 and can be inserted into the positioning hole 112 after reaching a proper position so that the positioning of the umbrella holder extension gun 11 In order to prevent erroneous operation of the control button 12, The needle bar 111 extends upward along the inner wall of the umbrella handle extension gun 11 and extends upwardly from the knurled hole (not shown) of the control button 12 121 can be inserted into the control button

12. When the control button 12 is operated, it is possible to prevent the erroneous operation by preferentially releasing the restriction of the knurl 111 with respect to the control button 12. Figs. Description of the Working Example of the Embodiment 1 After the umbrella handle extension gun 11 is collected, the umbrella handle extension gun 11 protects the umbrella handle and protects the umbrella 111 from being inserted into the knurled hole 121 The spring bead 104 is inserted into the positioning hole 112 in the lower portion of the protruding portion 114 to form the positioning. You can increase the grip length of the umbrella handle after pulling, making it convenient for people to grip The knurled knuckle 111 is moved down along the umbrella handle extension gun 11 so as to be controlled by the control button 12 by releasing the knurled knot hole 121. The umbrella handle extension gun 11 is moved downward The bottom cover 113 moves downward and the spring bead 104 moves away from the positioning hole 112 in the lower portion of the projection 114 and relatively moves upward and reaches the proper position, The position of the umbrella grip 11 is determined and the constant state is maintained without continuing to move down. Figs. 4 and 5 show the structure of the second embodiment of the present invention A protrusion 201 is provided on the outer wall of the umbrella handle 20 and the protrusion 201 is formed in a spiral shape along the outer wall of the umbrella handle 20, Extends to the lower portion of the umbrella handle 20 A spiral concave groove 211 is provided on the inner wall of the umbrella handle extension cylinder 21 and the concave groove 211 passes through the inner wall of the umbrella handle extension gun 21 and both ends of the concave groove 211 It is possible to form the correct position with respect to the umbrella handle extension gun 21 in the course of ascending and descending the umbrella handle extension gun 21 and prevent the umbrella handle 20 from being disengaged. When the control button 22 is not operated The umbrella pull extension gun 21 is turned up to the upper side of the umbrella handle 20 to protect the control button 22. Conversely, when operating the control button 22, the umbrella pull extension gun 21 is inserted into the umbrella handle 20, the control button is operated and the length of the umbrella handle is increased to make it easier for people to grasp. Figs. 6 and 7 are explanatory diagrams of the structure of Embodiment 3 according to the present invention. In Embodiment 3, The seat 33 is inserted into the umbrella handle 30. An umbrella handle extension key 31 is provided on the outside of the umbrella handle 30 to completely cover the umbrella handle 30 to form a protection against the umbrella handle 30. Generally, 31 includes a bottom cover 313 which is fixed to the lower portion of the umbrella handle 30 and has protruding portions 314 extending upward from both sides of the middle portion thereof. There are provided a number of positioning holes 312 to correspond to the position of the umbrella handle extension key 31 and to form a positioning action by being engaged with the spring bead 304. On both sides of the umbrella handle 30, A spring 303 is installed to support the spring bead 304 and the spring bead 304 is inserted into the hole 305 in the process of moving the umbrella handle extension gun 31 up and down. Slides up and down along the projection 314 under the support of the spring 303 And reaches the proper position and then inserts into the positioning hole 312 to realize the correct position of the umbrella handle extension gun 31. On both sides of the umbrella handle 30, there are two downwardly extending

extension grooves 301, The extension groove 301 is symmetrically distributed and a flange 302 is provided on the underside of the extension groove 301 to engage with the convex block 311 on the inner wall of the umbrella handle extension gun 31. On the inner wall of the umbrella handle 31 The raised convex block 311 is installed in correspondence with the position of the extension groove 301 to be engaged with the flange 302 on the extension groove 301. After the umbrella handle extension gun 31 is folded, The spring bead 304 is inserted into the positioning hole 312 at the lower portion of the protrusion 314 so that the spring bead 304 is inserted into the positioning hole 312 at the lower portion of the protrusion 314 Position. The length of the umbrella handle is increased by pulling the umbrella handle extension needle 31 and the spring bead 304 is also moved while being inserted into the positioning hole 312 at the upper portion of the protrusion 314, It should be understood that the above description is only a few examples of the present invention and should not be construed as limiting the scope of the present invention, All similar cases fall within the scope of protection of this invention.

And an inner sheet of an umbrella handle is provided inside, and an umbrella handle capable of adjusting the length by the operation of a control button,

A handle extension gun is installed on the handle,

The handle extension gun is fitted to the handle to wrap the handle,

And a connecting mechanism is provided between the handle extension gun and the handle.

The method according to claim 1,

Wherein the connecting mechanism includes a protrusion and a groove coupled to the protrusion, the protrusion being installed in the longitudinal direction of the handle on the outer wall of the handle or on the inner wall of the handle extension, the groove being formed on the inner wall of the handle extension gun Wherein the handle is provided on an outer wall of the handle, and the protrusion and the groove are fitted to each other.

3. The method of claim 2,

Wherein the projection is a convex block provided on an inner wall of the handle extension gun, the groove is an extension groove member provided on the outer wall of the handle, a flange is provided at an end of the extension groove member, and the flange extends to the bottom of the convex block Features an umbrella handle with adjustable length.

4. The method of claim 3,

Wherein the extending groove member is an elastic member.

4. The method of claim 3,

Characterized in that at least one of the convex block and the extension groove member is installed and arranged symmetrically on the axis of the handle extension gun and the handle.

The method according to claim 1,

Wherein an elastic support mechanism is provided on an outer wall of the handle, and a support mechanism supported by the elastic support mechanism is provided on an extension key of the handle.

7. The method of claim 6,

Wherein the elastic support mechanism is a spring and a spring bead, the spring is installed in a spring hole of the handle to support the spring bead, and a positioning lever is provided on the handle extension key, Characterized in that the handle is provided on the gun and can extend along the inner wall of the handle, on which at least one positioning hole is arranged on the positioning lever.

4. The method of claim 3,

Wherein the protruding portion is formed in a spiral structure that surrounds the outer wall of the handle, and a spiral groove is formed in an inner wall of the handle extension gun so as to be coupled with the protrusion of the helical structure.

9. The method of claim 8,

Wherein the spiral projection is only partially mounted on the outer wall of the handle.

4. The method of claim 3,

Wherein the projection is provided on an inner wall of the handle extension gun, and the groove coupled with the projection is provided on an outer wall of the handle.